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CONTROL PLANE IN CLOUD SECURITY: CONTROL PLANE VS. DATA PLANE

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What is a control plane?

A **control plane** is a digital component that manages the flow of data across the network. While the control plane is not responsible for the actual transfer of data, it serves as the digital control center, enabling organizations to orchestrate communication between different network elements and manage decision-making related to routing, configurations, resource allocation and policy enforcement.

In a cloud environment, the **cloud control plane** serves as the backbone of the network, overseeing and coordinating cloud infrastructure elements, including provisioning, configuration and management of resources such as VMs, storage, and networking.

Security of the control plane—and especially the cloud control plane—is of paramount importance to organizations. Findings from the **CrowdStrike 2024 Threat Hunting Report** revealed that the cloud control plane has become a key target for adversaries since once an adversary gains access, they have nearly complete control of the cloud infrastructure.

Control plane vs. data plane

There are two main components that manage data flow within the network: the control plane and the data plane.

While these components work together to ensure the smooth flow of data and operation of the network, they have distinct roles and manage separate functions.

The role of the control plane

The control plane acts as a network orchestrator. It creates routing tables, which determine the optimal path data should take during transfers and how to manage data traffic. Additional functions of the control plane include:

- Network provisioning
- Configurations
- Network policy creation and enforcement
- Network topology
- Security
- Load balancing

The role of the data plane

The data plane, sometimes called the forwarding plane, facilitates the actual transfer of data packets within the network—or how data gets from point A to point B based on control plane logic. This includes:

- Receiving and inspecting packets
- Queueing network packets during periods of high traffic
- Confirming safe delivery of the packet to the intended destination

Control plane vs. data plane in-depth

Understanding the distinctions between the control and data planes is crucial for optimizing network performance and ensuring effective security policies.

Plane	Control plane	Data plane
Function	Decides how data traffic is managed and routed across the network	Facilitates the movement of data from its origin to its destination
Purpose	Builds and manages the IP routing table, which establishes the logic by which data traffic will be managed, routed, and processed	Transfers data in accordance with the logic established by the control plane
Communication	Uses protocols to communicate between different systems (e.g., BGP, OSPF, IS-IS and SNMP)	Uses dedicated networks to communicate between systems (e.g., WiFi, cellular, satellite communications, virtualized networks, industrial control systems, or IoT)
Location	Cloud-based	Functions within the data processing area

How the control plane interacts with the data plane

The control plane and data plane work interdependently, with each component serving a distinct but complementary role for the safe operation of the network and transfer of data.

Here, we outline two key interactions between the control plane and data plane: policy enforcement and management efficiency.

Policy enforcement

The control plane determines how the data plane operates by executing and enforcing a variety of predefined policies.

For example, in a cloud environment, the control plane may specify access controls for certain users, services or both. As a result, the control plane can enforce which users can interact with what applications or resources and apply the rules consistently. This helps the organization maintain security and compliance, while also managing data flow and resource allocation.

Management efficiency

Though the control plane and data plane are connected, they are structured separately. This allows organizations to respond to changing demands and manage both planes more effectively; in a cloud environment, it also enables teams to scale cloud resources quickly and securely.

For instance, during a traffic spike, the control plane can dynamically update routing rules or reallocate resources to better manage peak network needs. The system can also implement new security protocols on-demand, enabling teams to protect the network from security threats as they arise.

Cloud control plane in modern security architecture

As noted above, a **cloud control plane** is the network component responsible for managing and orchestrating the cloud infrastructure. This includes:

- Provisioning
- Managing user permissions, including application programming interface (API) access
- Configuration and management of resources such as VMs and storage
- Multi-tenant resource isolation
- Networking

Benefits of control plane security in cloud detection and response

Control plane security plays a critical role in cloud detection and response, offering enhanced protection through real-time monitoring, automated policy enforcement, and optimized access controls.

- **Enhanced security**: Enhanced visibility into the control plane enables organizations to identify unusual behavior, unauthorized access, or configuration changes as they happen. This proactive approach ensures faster threat detection and response, minimizing the potential damage caused by security incidents and improving the overall security posture.
- **Automated policy management**: By automating the enforcement and management of security policies, the control plane reduces the reliance on manual processes, which are prone to human error.
- **Optimized access controls**: Strengthened user authentication and resource permission management are critical for securing multi-tenant environments and preventing unauthorized access. Control plane security enables granular access controls, ensuring that users and applications have the least amount of privilege required, which helps prevent unauthorized access and also limits an attacker's reach if the network is breached.

Security risks and challenges

Organizations face unique risks and challenges when securing the cloud control plane. Three of the most common are: access controls, misconfigurations and compliance.

Access control

The cloud control plane holds the metaphorical keys to the kingdom. This component is responsible for managing sensitive configurations and access permissions, making it one of the most vulnerable components of the cloud infrastructure and a prime target for attackers.

Since companies often grant employees more access and permissions than needed to perform their job functions, this can increase the risk of identity-based attacks within the control plane. Further, misconfigured access policies—or default access controls set by the cloud provider—are common issues that often escape security audits and can be exploited by a threat actor.

Misconfigurations

Misconfigurations are the number one vulnerability in a cloud environment. These can lead to overly permissive privileges on accounts, insufficient logging, and other security gaps that expose organizations to data breaches, cloud breaches, insider threats, and adversaries who leverage vulnerabilities to gain access to the data and network.

Misconfigurations are especially critical when it comes to the cloud control plane because this serves as the backbone of the network. Vulnerabilities here can enable the attacker to not only access the system but escalate the attack.

Compliance

Depending on their industry, companies may hold sensitive customer information, such as credit card numbers, social security numbers, addresses, and health information, in the cloud. Control planes, like any cloud component, must adhere to various regulations to protect such information. Misconfigurations or negligence can lead to security failures and result in fines or penalties, underscoring the importance of effective governance and compliance.

Best practices for the cloud control plane

Due to the critical nature of the cloud control plane, organizations should take steps to strengthen security. Here, we share some best practices to help overcome common cloud control plane challenges and risks.

User identity and access management

- Implement robust identity and access management (IAM) services native to your cloud platform to implement role-based, fine-grained access control to cloud resources.
- Require multi-factor authentication (MFA) for all users accessing the control plane. This adds an extra layer of security beyond passwords, mitigating the risk of unauthorized access.
- Leverage tools offered by cloud platform providers to integrate on-premises solutions like Active Directory with cloud-native IAM services; this can provide users with a seamless single sign-on (SSO) experience for cloud-hosted workloads.
- Adhere to the principle of least privilege when it comes to access rights to all cloud resources, including APIs. According to this principle, users are only allowed to access the data and cloud resources they need to perform their work. This can help limit an attacker's ability to advance in the network and escalate privileges once access is established.

Configuration

- Standardize and validate cloud resource configurations before deployment; regularly monitor for deviations from approved standards, which could indicate an attack.
- Incorporate cloud security posture management (CSPM) solutions into your architecture to monitor for misconfigurations that could creep into your cloud deployment.
- Apply the same security policies to cloud workload servers as any other server, and deny outbound connections initiated from any server that does not link to allowlisted endpoints. These policies should be applied across the board by implementing ingress and egress filtering and monitoring your cloud assets.

Monitoring

- Implement tools to monitor control plane activity in real time, detect unusual behavior, and generate alerts for potential security threats.
- Enable logging capabilities within the cloud infrastructure to gain full visibility into the network and quickly identify unusual activity. Turn on notifications within the log management platform to ensure you receive real-time alerts on unusual or suspicious activity.
- Monitor cloud assets and mitigate risks in a timely manner. Keep an eye on any new vulnerabilities or configuration changes which could increase risk.

Conclusion

Securing the control plane is critical to a strong cloud security strategy, particularly in complex multi-cloud and hybrid environments. As the central management point for cloud policies and permissions, the control plane demands stringent security measures to prevent unauthorized access and help ensure compliant operations across environments.

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